

CS 2XY3 Final Exam (Problems)

Time: 150 minutes. Total: 100 points. Answer all questions.

Question 1 (20 points) - Deadlocks

- (a) State all four necessary conditions for deadlock. (4 pts)
- (b) Given a resource allocation graph with processes P1-P4 and resources R1-R3 where: P1 holds R1 and requests R2; P2 holds R2 and requests R3; P3 holds R3 and requests R1; P4 requests R2. Is there a deadlock? Identify the cycle. (8 pts)
- (c) Describe the Banker's Algorithm. What information does it require? (8 pts)

Question 2 (20 points) - File Systems

- (a) Compare contiguous, linked, and indexed file allocation methods. Give one advantage and one disadvantage of each. (9 pts)
- (b) A file system uses 4-KB blocks and 4-byte block pointers. An inode has 12 direct pointers, 1 single-indirect, 1 double-indirect, and 1 triple-indirect pointer. What is the maximum file size? (11 pts)

Question 3 (20 points) - Virtual Memory

- (a) Explain the page replacement process when a page fault occurs. (5 pts)
- (b) Given reference string: 7,0,1,2,0,3,0,4,2,3,0,3,2 with 3 frames, calculate page faults for FIFO and LRU. (10 pts)
- (c) What is thrashing and what causes it? (5 pts)

Question 4 (20 points) - I/O and Disk

- (a) Calculate the average seek time for SCAN (elevator) disk scheduling given head position 53 and request queue: 98, 183, 37, 122, 14, 124, 65, 67 (moving toward 0 first). (10 pts)
- (b) What is DMA and why is it important for I/O performance? (5 pts)
- (c) Compare programmed I/O, interrupt-driven I/O, and DMA. (5 pts)

Question 5 (20 points) - Protection and Security

- (a) Explain the difference between authentication and authorization. (4 pts)
- (b) What is an access control matrix? How do access control lists (ACLs) and capabilities relate to it? (8 pts)
- (c) Describe two types of security threats (e.g., buffer overflow, Trojan horse) and one mitigation technique for each. (8 pts)